HRS Net: A Fusion model of HRNet and RESNet

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1. Introduction

Deep convolutional neural networks have attained the forefront in numerous computer vision applications, including image classification, object detection, semantic segmentation, and human pose estimation, among others, by achieving state-of-the-art performance. However, some of the previously known algorithms have a vanishing gradient problem which occurs when the value of the gradient becomes so small that it has an insignificant effect on the updation of parameters which hinders its ability to scale higher and implement larger layers.

HR Net is unique from the other algorithms, in a way that it maintains high-resolution representations throughout the process, by utilizing parallel multi-resolution subnetworks i.e. both down-sampling and up-sampling in parallel and repeated multi-scale fusions. Previous algorithms use the approach of down-sampling and then up-sampling resulting in the loss of the local information. Albeit computationally expensive, the parallel multi-resolution subnetworks utilized in HR Net help overcome this limitation and improve accuracy, which could have useful applications in varied fields across life. This provided us with a major stimulus behind choosing to implement this paper.

Applications of HR Net are not just limited to Human Pose estimation and can be applied to image classifications and semantic segmentations also. In the original paper [1], the algorithm has been used to perform pose estimation and semantic segmentation. The goal behind our project is to go a step further and train a model that can perform image classification by implementing a fusion model of HR Net and RESNet.

2. Related Work

The evolution of Convolutional Neural Network (CNN) architectures has laid the foundation for advanced image analysis and understanding. Foundational models such as AlexNet [2], VGGNet, ResNet [3], and Inception have introduced important concepts like deep layering, residual connections, and inception modules. Unlike older methods that use pyramid structures or encoder-decoder designs for tasks like semantic segmentation (for example, PSPNet, DeepLab series, U-Net [4]), the High-Resolution Network (HRNet) is notable for keeping high-resolution information throughout the network. This is different from models that gradually reduce image detail, which can lose important information for tasks like semantic segmentation.

In the area of human pose estimation, architectures like Convolutional Pose Machines, Stacked Hourglass Networks, and SimpleBaseline have been created to accurately capture body poses and spatial relationships. These models often involve iterative refinement and multi-stage processing to meet the specific demands of pose estimation tasks. Efficiency and speed are crucial, especially for tasks where quick responses are needed. Researchers discuss ways to make models more efficient, like using simpler designs, cutting unnecessary parts, simplifying calculations, and using faster methods for processing data. These efforts aim to ensure that advanced models like HRNet can be used in real-world situations where quick analysis is essential.

While HRNet is great at estimating human poses, it's also useful for other tasks like figuring out what objects are in an image, dividing images into meaningful parts, and classifying images. Its flexibility and effectiveness in different jobs show that it's a valuable tool in computer vision, able to tackle a range of challenges by keeping detailed representations. Compared to models made for specific tasks, HRNet's broad abilities highlight its importance as a flexible and powerful tool in the changing world of computer vision.

3. Baseline Method

The High-Resolution Network (HRNet) is a novel architecture designed to maintain high-resolution representations through the entire network, contrasting with traditional convolutional neural networks that reduce the resolution of feature maps early in the network. This characteristic makes HRNet particularly well-suited for tasks requiring detailed spatial information, such as image segmentation, object detection, and even pose estimation.

The network is divided into four stages. The first stage consists of a single high-resolution branch. Subsequent stages introduce branches of lower resolutions, each doubling the number of branches from the previous stage until the final configuration is reached. At the end of each stage, features from different resolutions are fused before transitioning to the next. This fusion happens through a series of up-sampling and down-sampling operations, ensuring that each branch receives information from all other branches. This mechanism allows HRNet to reinforce high-resolution features with context from lower resolutions throughout its depth.[1]

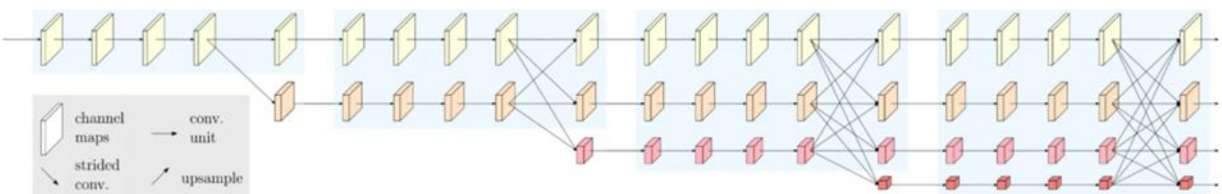

Fig 1: Four stages of HRNet.

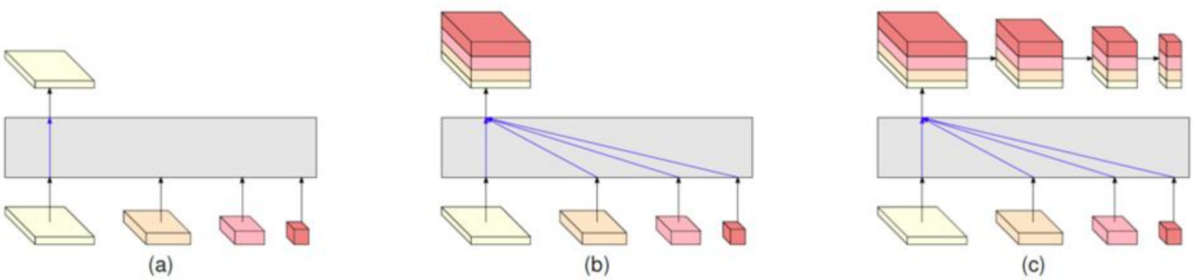

Fig 2: Comparative Architectural Diagrams of HRNet Variants. Source: [Microsoft Research Blog]

ResNet, one of the popular neural network architectures excels in creating deep networks through residual connections, optimizing for feature abstraction at reduced spatial resolutions, making it versatile across a wide range of tasks like image classification [5]. HRNet, on the other hand, maintains high-resolution representations throughout the network, focusing on capturing detailed spatial information, which is particularly beneficial for tasks requiring fine-grained detail, such as semantic segmentation [6] and human pose estimation [7][8].

4. Proposed Method

The main goal of our project is to carefully recreate the HRNet architecture using PyTorch, ensuring an accurate representation of its structure and capabilities. This foundational implementation will serve as the basis for our subsequent objectives. Once the HRNet backbone is established, we plan to enhance it with various classifiers to make it adaptable for different computer vision tasks such as object detection, image classification, human pose estimation, and segmentation realizing the true power of such a multi-modal model in visual recognition challenges.

Simultaneously, we aim to create a fusion model by combining HRNet with ResNet, specifically tailored for image classification tasks. This hybrid architecture aims to leverage the fine-grained feature extraction of HRNet and the stability of ResNet's residual connections to achieve a powerful model capable of capturing intricate details while maintaining robustness. We are curious to see the outcome of this fusion model as both these models individually have succeeded and are widely used. We will train and evaluate HRNet using the ADE20K dataset for high-resolution segmentation tasks and evaluate the models on ImageNet datasets for a comprehensive assessment of their performance in image classification.

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